

AgentPay × Substreams

Architecture Update for The Graph

From: Shawn Lippert, AgentPay | May 19, 2026

Re: Substreams integration status + architecture questions

What We Shipped

We just crossed a major milestone: the AgentPay Substreams package is **compiled, packed, and streaming live on Base mainnet** as of today.

- **Package:** agentpay-substreams-v0.1.0.spkg
- **Network:** base-mainnet via mainnet.eth.streamingfast.io:443
- **Status:** Live — 8+ blocks processed, usage confirmed via StreamingFast dashboard
- **API Key:** server_903d0... (StreamingFast / The Graph Market)

Pipeline Architecture

Base L2 Blocks



[map_usdc_transfers]

- Watches Base USDC: 0x833589fCD6eDb6E08f4c7C32D4f71b54bdA02913
- Filters on TransferWithAuthorization topic (EIP-3009)
- Topic: 0x98de503528ee59b575ef0c0a2576a82112635b28b9da4cae03c8b0b0ec06b8e3
- Extracts: from, to, value, nonce, block_num, timestamp, tx_hash



[map_confirmed_settlements]

- Filters to transfers TO AgentPay facilitator wallet
- Emits SettlementEvent (payment_id derived from EIP-3009 nonce)
- Hot path: facilitator releases service within the same block



■■■■ [store_agent_counters] StoreAddInt64 – accumulates across blocks



Keys: total, success, refunded, diversity per agent



■■■■ [store_agent_reputation] StoreSetProto<AgentReputation>



Score: 60% success rate + 25% diversity + 15% recency



■■■■ [map_analytics_events]

AnalyticsEvent → Clickhouse sink

Fields: agent, counterparty, amount_usdc, score_ppm, city, tx_hash

Why This Matters for The Graph Integration

The core problem we were solving: our old facilitator used a polling loop — `eth_getLogs` every 2 seconds. At scale:

- **Latency:** 2–4 second confirmation lag on every payment
- **Cost:** ~\$0.002/req × high frequency = real ongoing expense
- **Fragility:** RPC provider outages = missed or delayed settlements

With Substreams streaming directly from Firehose:

- **Latency:** Sub-block — confirmed within the same block the tx lands
 - **Cost:** One persistent gRPC stream, zero polling overhead
 - **Reliability:** StreamingFast infrastructure, not a single RPC endpoint
 - **Scale:** GraphQL queries hit the subgraph, not our facilitator — server load stays flat
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Questions for You

1. x402GrantRegistry

Do you have the contract address + `startBlock` for the grant registry on Base? We want to index it as a fifth module so grant-funded settlements are distinguishable from standard payments in the analytics stream.

2. Clickhouse Sink

We've built the `AnalyticsEvent` proto to match the Clickhouse sink schema pattern you mentioned. Is there a recommended `substreams-sink-clickhouse` version for Base mainnet, or are you running a custom fork?

3. Subgraph Deployment Path

Once we have the grant registry address, the plan is:

- Deploy the subgraph to The Graph Network (decentralized)
- Expose a GraphQL endpoint so external agents can query `agentReputation(wallet: "0x...")`
- Wire the reputation score back into our facilitator's credit scoring layer

Is there a preferred path for getting an `AgentPay` subgraph onto the decentralized network, or should we start on the hosted service first?

What We Need from You

Just two things to unblock the next phase — everything else is built and running:

1. x402GrantRegistry contract address + `startBlock` on Base
2. Your preferred Clickhouse sink version / config for Base mainnet

The `.spkg` is packed, the pipeline is streaming, and the facilitator client is wired up to consume the gRPC stream. Happy to share the repo directly or coordinate a PR review whenever works for you.

Shawn Lippert | AgentPay Team | x402-agent-pay.com | [@shawnhvac](https://twitter.com/shawnhvac)

GitHub: github.com/shawnhvac/x402 | 95b Havasupai St, Grand Canyon, AZ 86023